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Measurable Functions

Definition 1.0.1

We endow $\overline{\mathbb{R}}$ with a topology called the order topology in the following manner. For each $a \in \mathbb{R}$ let

$$L_a = \overline{\mathbb{R}} \cap \{x : x < a\} = [-\infty, a) \quad \text{and} \quad R_a = \overline{\mathbb{R}} \cap \{x : x > a\} = (a, \infty].$$

The collection $\mathcal{S} = \{L_a : a \in \mathbb{R}\} \cup \{R_a : a \in \mathbb{R}\}$ is taken as a subbasis for this topology. A basis for the topology is given by

$$\mathcal{S} \cup \{R_a \cap L_b : a, b \in \mathbb{R}, a < b\}.$$

Observe that the topology on $\mathbb{R}$ induced by the order topology on $\overline{\mathbb{R}}$ is precisely the usual topology on $\mathbb{R}$.

Definition 1.0.2 ► Measurable Functions

- Suppose $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are measure spaces. A mapping $f : X \rightarrow Y$ is called **measurable with respect to $\mathcal{M}$ and $\mathcal{N}$** if $f^{-1}(E) \in \mathcal{M}$ whenever $E \in \mathcal{N}$. If there is no danger of confusion, reference to $\mathcal{M}$ and $\mathcal{N}$ will be omitted, and we will simply use the term “measurable mapping.”